PRODUCT REQUIREMENTS DOCUMENT • V1.0

AIOps Sentinel: Autonomous SRE Engine

Closed-Loop Incident Detection, Root Cause Analysis, and Self-Healing Automation

AUTHOR	TARGET RELEASE	TARGET SLA	DOCUMENT STATE
SRE & AI Platform Team	Q1 Production Milestone	99.99% Availability	Ready for Engineering

1. Executive Summary & Vision

Modern distributed architectures suffer from high Mean Time to Resolution (MTTR) averaging 35 to 55 minutes per incident, primarily consumed by manual log inspection and context synthesis. AIOps Sentinel acts as an autonomous reliability agent that detects pre-crash telemetry anomalies using machine learning, synthesizes root causes with LLM-driven retrieval against runbooks, and validates corrective actions inside an ephemeral sandbox before applying verified live remediations.

Core Problem Statement

Engineers encounter alert fatigue from redundant notifications while critical silent failures (e.g., resource starvation, connection pool leaks) progress unaddressed until downtime occurs. Current tools alert humans passively; AIOps Sentinel transitions operational reliability from human-mediated troubleshooting to machine-verified autonomous self-healing.

2. Target Personas & Operational Needs

DevOps & SRE Engineers (Primary)

Require zero alert storms during off-hours, trusted execution guardrails, deterministic command execution logs, and automated incident post-mortems formatted in Markdown.

Application Developers (Secondary)

Require quick zero-touch SDK integration (Node.js/Python), automated GitHub Pull Requests for code-level crash bugs, and real-time slack notifications for active issues.

3. Key Performance Indicators (KPI Targets)

≤ 15s	< 3 min	> 94%	100%
MEAN TIME TO DETECT	AUTONOMOUS MTTR	DETECTION PRECISION	SANDBOX VERIFIED

4. Functional Requirements Matrix

Req ID	Component	Functional Description	Priority
FR-01	Telemetry Ingestion	Zero-lag buffering of metrics (CPU, RAM, I/O) and container logs via Redis Streams and FastAPI.	P0 (Blocker)
FR-02	ML Anomaly Engine	Unsupervised Isolation Forest + DistilBERT classifier evaluating sliding windows in ≤ 100ms.	P0 (Blocker)
FR-03	Generative RCA	RAG retrieval of system topology and runbooks to synthesize structured JSON root-cause diagnostics.	P0 (Blocker)
FR-04	Sandbox Testing	Ephemeral Docker validation of shell scripts with maximum 3 reflection and self-correction cycles.	P0 (Blocker)
FR-05	Action Dispatcher	Dual execution support: Autonomous mode via Docker/K8s APIs or Safe mode via Slack interactive cards.	P1 (High)

5. Incident Lifecycle & Verification Workflow



Figure 1.0: End-to-end incident lifecycle from anomalous log detection to verified container remediation.

6. Operational Modes & Governance Guardrails

The platform enforces strict blast-radius controls. Clients select between **Safe Mode**, which delivers an enriched interactive Slack card requiring one-click manual approval, and **Autonomous Mode**, reserved for non-destructive actions such as memory cache purges, container reboots, and horizontal replica scaling. Destructive operations (disk formatting, database drops) are strictly intercepted by an immutable Abstract Syntax Tree command filter.